

FREE

Resolve Your Inflammation With This One Highly Unique Form of Lactoferrin

Not all lactoferrin is created equal!



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One of the most promising natural resolvers of inflammation is a milk-derived protein known as lactoferrin.

As we will see momentarily, however, **not all lactoferrin is created equal**.

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Lactoferrin is naturally present in milk, saliva, vaginal secretions, semen, mucosal linings, and in polymorphonuclear leukocytes — neutrophils, eosinophils, and basophils — involved in the immune response.

The immune system seems to release it to help bring iron into cells. Iron can feed some pathogens, so an infection drives interleukin-6 (IL-6) to drive lactoferrin release from immune cells and the production of the iron storage protein ferritin inside most cells. The lactoferrin helps bring the iron from the blood into the cells where it gets stored in the ferritin.

Lactoferrin is especially unique in its ability to pick up iron under acidic conditions such as the local sites of infections. It can therefore pick up free iron in infected tissue that the more everyday iron chaperone transferrin would not be able to pick up.

Since lactoferrin accomplishes the mission of keeping iron safely hidden away from pathogens, it sends a “mission accomplished” signal to the immune system by suppressing the continued secretion of IL-6.

This means that for people who are chronically inflamed, lactoferrin — at least, the right type of lactoferrin — can help resolve the inflammation by bringing IL-6 back down to normal.

Each molecule of lactoferrin can bind to two iron atoms, and each 100 milligrams of lactoferrin can bind to 3.57 milligrams of iron. The main *dietary* source of lactoferrin is milk. (Especially raw milk, since pasteurization destroys 65% of the lactoferrin). In milk, 15-20% of lactoferrin’s iron-binding sites are bound to iron. The portion bound to iron is known as holo-lactoferrin or iron-saturated lactoferrin, while the portion that is not bound to iron is called apo-lactoferrin.

While all natural lactoferrin is a mix of the two, nearly all supplements on the US market are 100% apo-lactoferrin.

In [one Italian clinical trial](#), 100 milligrams twice a day of 20-30% iron-saturated lactoferrin taken just prior to meals improved anemia in women with various forms of chronic inflammation just as effectively as an iron supplement. This was striking because the lactoferrin contained less than one milligram of iron whereas the iron supplement provided 105 milligrams per day of iron as ferrous sulfate. Despite having so little iron, lactoferrin increased serum iron by 48%. It accomplished this by cutting IL-6 in half, which allowed the iron to escape the grasp of ferritin, thereby allowing it to be mobilized for circulation throughout the body and also for hemoglobin production.

That trial was not randomized, but a [randomized controlled Egyptian trial](#) in adults with chronic kidney disease and anemia and another in [children](#) found similarly that the improvement in hemoglobin was the same or better with lactoferrin than iron and that the lactoferrin reduced IL-6 levels.

I have reviewed the literature more broadly in the appendix. ¹

While the 20-30% iron-saturated lactoferrin has not been studied head-to-head against apo-lactoferrin, iron-saturated lactoferrin is more resistant to digestion, has a different profile of antimicrobial activity, is more reflective of what is available from natural foods, and is what has been studied for IL-6 lowering and therefore what we should base our conclusions on.

The smaller body of literature using apolactoferrin does not show clear reductions in IL-6.

Thus, the natural, partially iron-saturated form of lactoferrin is what should be used to try to resolve sticky inflammation.

Until recently, the only lactoferrin supplement on the market was an Italian product, [Lattoglobina](#).

However, two brands have recently started selling such a product in the US:

- [Double Tree](#) sells a natural partially iron-saturated lactoferrin that appears to be purified from dairy products using a patented process with high purity.
- The Lactoferrin Co sells 95% pure partially iron-saturated lactoferrin isolated from grass-fed milk, as a [powder](#) or as enterically coated [capsules](#).

The need for enteric coating is unclear and has not been clinically studied. Taking the lactoferrin on an empty stomach can prevent the stomach from becoming acidic enough to degrade it, and the successful anemia trials approximated this by having the lactoferrin taken just prior to the meals. It is likely the Double Tree product is more pure for those sensitive to dairy, but neither product is dairy-free. It is worthwhile to experiment with the three products from the two brands to compare their utility since their differences lie in gray areas.

100 milligrams twice a day cut IL-6 in half, so it might be that 400 milligrams per day in 2-4 divided doses could completely normalize IL-6. While this has not been studied much, it is worth trying 200-400 milligrams of natural iron-saturated lactoferrin per

day in 2-4 divided doses, taken before meals or on an empty stomach, to help resolve stubborn inflammation, especially if your IL-6 is high.

While milk naturally contains lactoferrin, and while the early milk meant for a newborn known as colostrum is especially enriched in it, a capsule of colostrum only contains about 2.6 milligrams of lactoferrin and raw milk (pasteurization destroys over half of the lactoferrin) probably only has around 10 milligrams per cup. Thus, colostrum is a good source if you are drinking it in volume like a newborn would and raw milk is a good source if it makes up most of your calories, but these are not good sources to obtain 200 to 400 milligrams of lactoferrin per day as a *supplement* to your existing diet.

On the other hand, while whey protein manufacturers do not measure and report the lactoferrin contents of their products and there is about two-fold variation between products according to the literature, we can say that one to two 20-gram scoops of [whey protein](#) will provide about 200 milligrams of natural partially iron-saturated lactoferrin, and you can use two to four scoops to obtain 400 milligrams.

The supplements allow you more optionality for your diet since you don't need to occupy a large portion of your protein requirement and a substantial portion of your caloric load with a protein powder.

I have updated several guides, ebooks, and articles that had only linked to the Italian product to now include the Double Tree and Lactoferrin Co products:

- [The Vitamins and Minerals 101 Cliff Notes](#)
- [Vitamins and Minerals 101: Iron](#)
- [The Iron Deficiency Protocol](#)
- [The Crohn's Protocol](#)
- [The Science Behind the Crohn's Protocol](#)
- [The COVID Guide](#)
- [Is Lactoferrin Chelating Your Iron?](#)

The Iron Deficiency and Crohn's protocols and the COVID Guide also include dairy-free options for lowering IL-6.

1 Appendix: How Robust Is the Conclusion?

How robust is the conclusion about natural, partially iron-saturated lactoferrin lowering IL-6 and apolactoferrin failing to do so?

The first anemia trial mentioned was not randomized or blinded, so it suffers from the potential of various biases, yet it makes physiological sense and paints a coherent picture across mechanisms and outcomes.

A [randomized and controlled but not blinded trial](#) in 70 patients with chronic kidney disease, high IL-6, and anemia, showed that the addition of 200 milligrams of lactoferrin per day from the Egyptian product [Provan](#), which is [30% iron-saturated](#), to the background

treatment of agents that stimulate hemoglobin synthesis, lowered IL-6 by 71% from 31 to 9 picograms per milliliter.

In [92 obese children with IBD](#) and iron deficiency anemia randomized to three months of 6 milligrams per kilogram bodyweight per day iron as ferrous sulfate or 100 milligrams of lactoferrin per day, hemoglobin improved better on lactoferrin than on iron, and lactoferrin decreased IL-6 went down by 34%. This used a product called Pravotin, which is [30% iron-saturated](#).

A study published in [Nature Scientific Reports](#) earlier this year was randomized but not blinded found similar results in children with chronic kidney disease: 33% lower IL-6 and improvement in anemia that was identical to iron supplementation. The dose of lactoferrin was 100 milligrams per day and it was given for three months and compared to 50 milligrams of intravenous iron dextran three times per week. Nothing is stated about the type of lactoferrin used or its sourcing, but this study is also Egyptian, raising to me the possibility it used either the Provan or Pravotin product.

[Another paper out of Egypt](#) described their lactoferrin as “regular,” whatever that means, and in a 3-month randomized but not blinded trial in obese children with iron deficiency anemia, the improvement in hemoglobin was superior with 100 milligrams of lactoferrin taken 15 minutes before meals compared to 6 milligrams per kilogram iron as ferrous hydroxide polymaltose. The IL-6 went down 68% from baseline in the lactoferrin group but remained insanely high (55 pg/mL), while it more than doubled in the iron group. IL-6 increasing with iron supplementation is not surprising because IL-6 would help raise ferritin to sequester the excess iron, but the levels reported seem way too high and probably result from a calculation error. The study is not listed on pubmed, adding further question about it. If the results are real, I similarly wonder if it used either the Provan or Pravotin product.

In [21 pregnant women](#) with iron deficiency anemia at risk for preterm delivery, 100 milligrams of lactoferrin taken orally twice a day before meals lowered IL-6 in cervicovaginal fluid by 43%. The authors made the [embarrassing mistake](#) of referring to their lactoferrin throughout the entire paper as recombinant human lactoferrin when it was in fact bovine lactoferrin. [Another paper](#) found dramatic resolution of high IL-6 in cervicovaginal fluid when administered vaginally to women at low risk of preterm labor for shortened cervix. These papers are from Italy, and my suspicion is they are using partially iron-saturated lactoferrin like the original Italian anemia paper.

An American study using a genetically engineered rice-derived product in [older women](#) showed no impact on IL-6, but their IL-6 was not elevated and it is not clear whether this product was all the apo form or not.

A study using [Jarrow apolactoferrin](#) claimed to show it lowered IL-6 in diabetics, but the study claims to have randomized the diabetics to lactoferrin and control, yet only shows IL-6 before and after the lactoferrin compared to the healthy controls, which shows that the product did not do what the authors wanted it to, so they chose the most misleading way to present it in the best light, which suggests that apolactoferrin does not lower IL-6.

A double-blind, placebo-controlled Spanish study in [preterm infants](#) showed that 150 milligrams per kilogram bodyweight per day lactoferrin had no effect on IL-6 over four weeks, but IL-6 declined to normal in both groups during the four weeks, suggesting there was little opportunity for the supplement to cause additional lowering beyond that which occurred in the placebo group. They used the Italian partially iron-saturated lactoferrin used in the Italian anemia paper.

Overall it appears the following is true:

- IL-6 levels need to be elevated for anything to reduce them (obviously) so studies where IL-6 is not elevated do not show good results regardless of what product is used.
- There is a general sense across studies that 200 milligrams per day causes a larger decrease in IL-6 than 100 milligrams per day.
- All of the successful studies either use a partially iron-saturated product or are conducted in countries like Italy or Egypt where every other study that does report the specific product reports it as partially iron-saturated.
- Two studies using apolactoferrin are not successful.

No studies compare apolactoferrin and partially iron-saturated lactoferrin head to head, but analysis of the existing literature makes it a reasonably supported hypothesis that partially iron-saturated lactoferrin is superior to apolactoferrin for this purpose. Since this is the natural form and there is a limited mechanistic basis for explaining why it would be superior, everything points towards using a partially iron-saturated product as a first choice with a higher probability of success..

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Discussion about this post

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Frank

3h

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♥

Liked by Chris Masterjohn, PhD

Thanks for the post. I see lactoferrin helps with type 1 diabetes, too.

♥

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SHARE



Brad n

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Great topic. Thanks. The italian product shared contains titanium dioxide so other options may be better. I looked at double wood one and saw this comment on Amazon reviews

"I was so hopeful to raise my ferritin with this. Please be very CAUTIOUS if that’s your goal! My ferritin was at 31. After 3 weeks of taking this... it’s 21. Though my iron profile looks good (and did before starting lactoferrin), it obviously pulls from ferritin. Definitely not taking it again."

So be careful if you have low ferritin.

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2 years ago my ferritin was 320(and total iron 190). I researched then and I don't remember what I found about lactoferrin but I started one from Jarrow(apolactoferrin) and few months later my ferritin normalized to 40(total iron 90). Now I don't remember if I was taking anything else at that time. Few months later I started again and checked - ferritin was 15(total iron 125) so I stopped it immediately.

Now my goal is to increase ferritin to 70-100 as I was reading on STTM. Planning to add iron bisglycinate/Vitamin C with apolactoferrin to increase my ferritin. Goal would be to increase ferritin without increasing iron.

If anyone has ideas please share.

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2 replies by Chris Masterjohn, PhD and others

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